PROJECT TITLE

THE TECH HIRING PARADOX

(Uncovering the Skill Gap and Experience Barrier Behind Modern Tech Unemployment)

An End-to-End Data Science Project Using API-Sourced Job Market Data

1. Project Objective

1.1 Business Problem

The objective of this project is to collect and transform publicly available tech job posting data into structured, decision-ready information through API-based data collection, data cleaning, exploratory data analysis, machine learning, and interactive visualization.

The project aims to identify trends, patterns, relationships, and anomalies in job postings, salary levels, required skills, and hiring activity across two structurally different segments of the tech job market — specialized roles (Data Scientist, AI Engineer, Cloud Engineer, Machine Learning Engineer) and legacy roles (Software Developer, PHP Developer, IT Support, Web Developer) — build a predictive model to classify postings into these segments, and communicate actionable insights through an interactive Power BI dashboard.

1.2 Main Business Question

Is tech unemployment driven primarily by a shortage of job openings, or by a structural mismatch between the skills companies demand and the skills most job seekers possess?

1.3 Objectives

- Collect 1,000+ job posting records from the Indian tech job market.
- Clean and preprocess the data.
- Perform exploratory data analysis across salary, skills, volume, and time.
- Identify relationships between job tier, salary, skill requirements, and posting trends.
- Build machine-learning models to classify postings as specialized or legacy.
- Compare model performance.
- Create an interactive Power BI dashboard.
- Generate actionable business insights and recommendations.

2. Data Collection Method

2.1 Data Source

Job posting data was originally planned to be scraped directly from Naukri.com. However, Naukri renders its job listings via JavaScript, and a Windows Application Control security policy on the development machine blocked the browser-automation tools (Selenium ChromeDriver) required to render such pages. The project pivoted to the Adzuna Jobs API — a public, documented, India-specific job search API that requires no login and is explicitly permitted for this kind of educational data collection.

2.2 Data Collection Technique

Python was used to collect job posting data from the Adzuna API using HTTP GET requests via the requests library. Eight keyword searches were issued — four representing specialized roles and four representing legacy roles — with results paginated at 50 records per page across 5 pages per keyword. Each response was parsed as JSON, and relevant fields were extracted directly from the API's structured response (no HTML parsing was needed, since Adzuna returns clean JSON).

2.3 Technology Used

- Python
- Requests
- Adzuna Jobs API (JSON)
- Pandas
- NumPy

2.4 Data Collection Workflow

Adzuna Jobs API (India) → Python Requests → JSON Response → Field Extraction → Per-Keyword Tier Tagging → Duplicate Checking → 2,000 Raw Movie... Job Records → Raw CSV Dataset

The source consisted of eight keyword searches × up to 250 records each = 2,000 raw records collected (4 specialized-role keywords + 4 legacy-role keywords).

2.5 Data Description

Raw dataset contains approximately: 2,000 rows × 11 columns.

Available useful fields included:

Column	Purpose
job_title	Title of the job posting
company	Hiring company name
location	Job location
salary_min / salary_max	Disclosed salary range (where available)
category	Adzuna's broad job category
contract_type	Permanent / contract (where disclosed)
created	Posting date
description	Job description text (used for skill extraction)
job_url	Link to the original posting
search_keyword / tier_label	Which of the 8 searches produced this row, and its assigned tier

Missing Fields

The Adzuna API did not provide usable values for applicant count, education requirement, or company size for the large majority of records, and truncates the description field to roughly 60–90 words. Contract type was disclosed for only 6.5% of postings. These limitations were not artificially filled in — fields were either excluded or explicitly flagged as "Not disclosed" / "Not specified" to maintain data reliability, rather than inventing values.

3. Data Cleaning & Preprocessing

3.1 Data Cleaning Objective

The raw job posting dataset was cleaned and transformed into a structured, analysis-ready dataset. The cleaning process focused on handling missing values, removing duplicate records, standardizing text and location fields, and engineering skill-based features from unstructured description text.

3.2 Cleaning Steps

1. Loaded the raw CSV dataset (2,000 rows)

2. Checked dataset dimensions and data types

3. Checked and handled missing values (fillna, not string-cast NaN, to avoid a silent pandas bug that let blank rows slip through undetected)

4. Removed 247 duplicate records (matched on title, company, and location)

5. Dropped 43 rows with missing job title or company

6. Standardized location field (flagged generic "India"/blank entries as "Not specified / Remote")

7. Converted salary fields to numeric and computed salary_avg

8. Flagged salary_disclosed as its own feature rather than imputing missing salary

9. Parsed posting dates

10. Extracted skill keywords from description text into specialized and legacy skill counts

11. Computed description_length (word count) as a feature

12. Encoded search_keyword numerically for potential use as a categorical feature

13. Filled missing contract_type with an explicit "Not specified" label

14. Sorted the final dataset by tier and keyword, with disclosed-salary rows surfaced first

3.3 Missing Value Handling

The Adzuna API did not provide usable values for applicant count, education level, or company size for the large majority of postings, and salary was disclosed for only 32.3% of records. These fields were not filled with artificial or estimated values. Salary non-disclosure was instead treated as a meaningful data point in its own right (see Section 4.5), and rows lacking essential identifying fields (title, company) were removed rather than retained with placeholder text.

3.4 Feature Engineering

Column	Purpose
specialized_skill_count	Count of modern/specialized skill keywords found in description (Python, AWS, ML, Cloud, etc.)
legacy_skill_count	Count of legacy skill keywords found in description (PHP, JavaScript, IT Support, etc.)
skill_count	Total skill keywords found (sum of the above)
description_length	Word count of posting description
salary_avg	Mean of salary_min and salary_max
salary_disclosed	Whether a salary figure was provided at all
search_keyword_encoded	Numeric encoding of which keyword search produced the row

After cleaning and preprocessing, the raw job posting data was transformed into a structured dataset of 1,710 records suitable for exploratory data analysis, machine learning, and Power BI visualization.

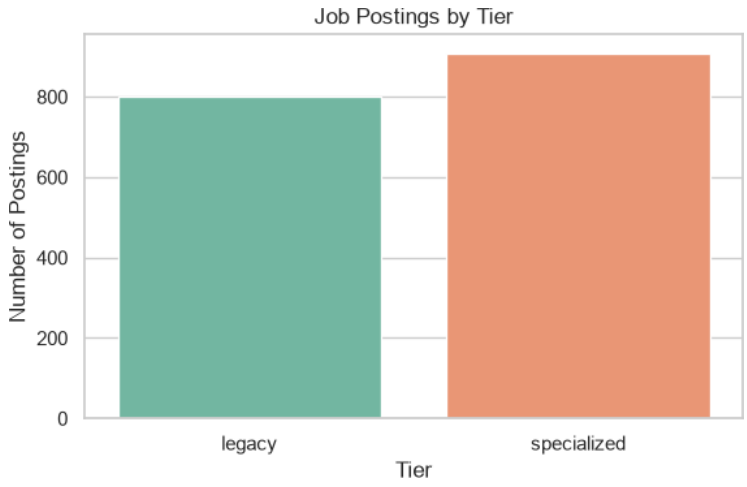
4. Exploratory Data Analysis (EDA) Findings

The cleaned job posting dataset was analyzed to understand tier volume, salary, skill requirements, hiring trends, and feature relationships.

4.1 : Descriptive Analysis

The descriptive analysis provides an overall understanding of the 1,710 job postings in the dataset, focusing on tier distribution, salary, and skill counts.

4.1.1. Job Postings by Tier

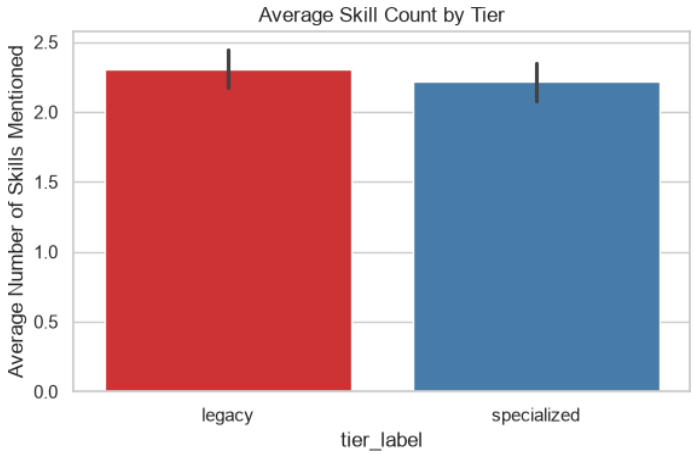


Job postings by tier.

Key observation: Specialized postings (909, 53.2%) and legacy postings (801, 46.8%) are nearly evenly split.

Key Insight: If tech unemployment were purely a matter of too few jobs overall, one tier would be expected to dominate the market by a wide margin. Instead, roughly equal volumes exist in both — suggesting the problem is not job scarcity but which jobs are available and what they require.

4.1.2. Average Skill Count by Tier



Average skill count by tier.

Key observation: Specialized postings average 2.22 skills per posting; legacy postings average 2.30 — nearly identical.

Key Insight: Raw skill count does not distinguish the two tiers. This is largely because "AI" has become a widely-used buzzword appearing even in many legacy-tagged postings, diluting skill count as a clean separator (see Section 4.2 for why skill identity still separates the tiers cleanly).

4.2 : Business Question Analysis

The business question analysis examines the relationships between job tier, salary, skill vocabulary, and posting trends across the 1,710 postings in the dataset.

4.2.1 : Tier vs Salary

“Do legacy roles pay more than specialized roles, or is it the other way around?”



Average disclosed salary by tier.

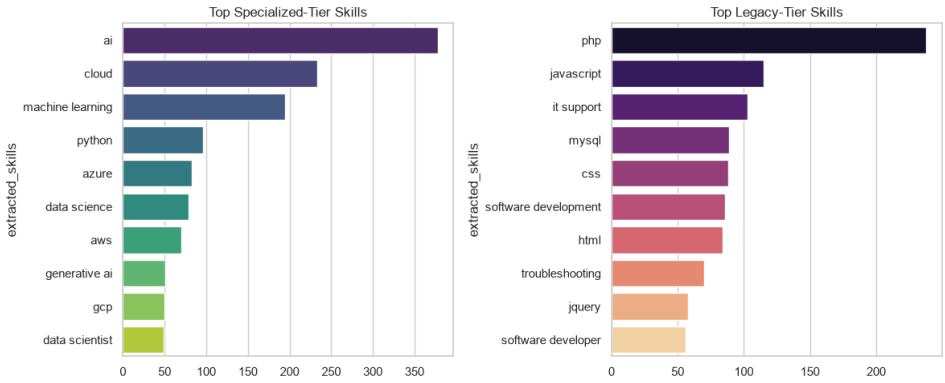
Metric	Specialized	Legacy	Gap
Average Salary	₹14.79 L	₹6.56 L	+₹8.23 L (+125.6%)

Key observation: Among the 552 postings (32.3%) that disclose a salary figure, specialized roles pay more than double the average legacy role.

Key Insight: This is the strongest single piece of evidence for the hiring-paradox thesis: companies are willing to pay a substantial premium for specialized skills, while legacy/generalist roles remain comparatively low-paying.

4.2.2 : Skill Vocabulary by Tier

"Which specific skills dominate each tier?"



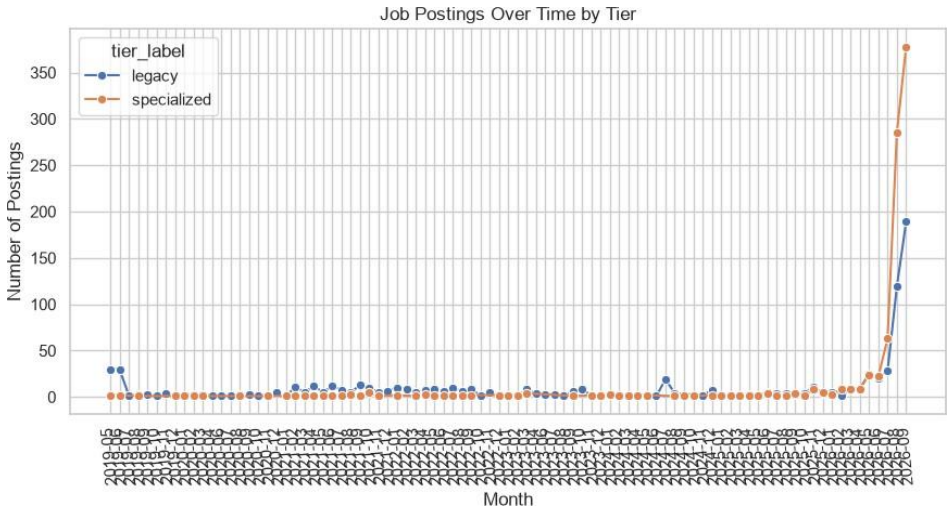
Top 10 skills mentioned in specialized-tier vs. legacy-tier postings.

Key observation: There is zero overlap between the top-10 skills of each tier. Specialized postings are dominated by ai, cloud, machine learning, python, and azure; legacy postings are dominated by php, javascript, it support, mysql, and css.

Key Insight: The tiers are not distinguished by how many skills are listed, but by which specific skills appear. Skill identity — not skill count — is what separates the market into two segments.

4.2.3 : Hiring Trend Over Time

“Is specialized hiring growing faster than legacy hiring?”



Job postings over time by tier.

Key observation: Comparing the first half to the second half of the data collection window, specialized postings grew by +82.3% while legacy postings fell by −49.7%.

Key Insight: This is the most direct evidence of an active shift in the market: employers are visibly reallocating hiring activity away from generalist roles and toward specialized ones. (Note: this reflects a short data-collection window and should be read as a within-period trend, not a confirmed multi-year pattern.)

4.2.4 : Salary Transparency by Tier

“How transparent is the market about salary, and does it differ by tier?”



Salary disclosure rate by tier.

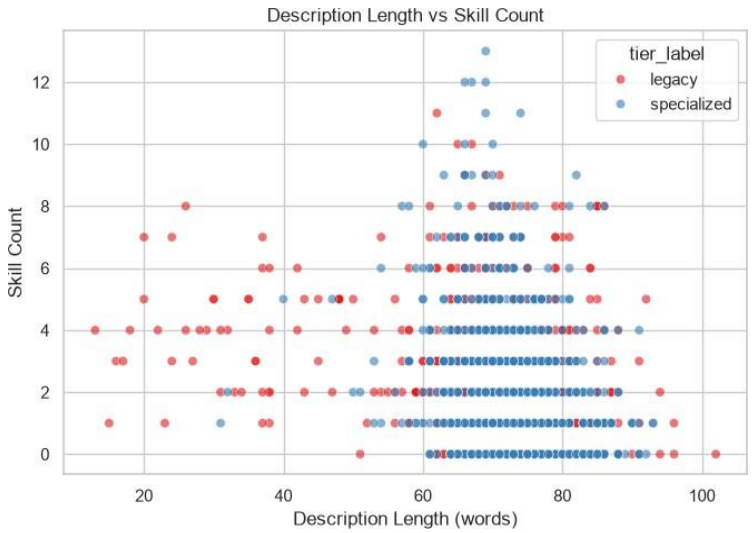
Key observation: Only 32.3% of all postings disclose a salary figure overall. Legacy roles disclose salary far more often (50.8%) than specialized roles (15.9%).

Key Insight: Specialized/premium roles appear to be negotiated more privately, while standard legacy roles are advertised with fixed, transparent pay bands — the reverse of what might be expected.

4.3: Relationship & Anomaly Analysis

4.3.1: Description Length vs Skill Count

"Do longer, more detailed job descriptions tend to list more skills?"

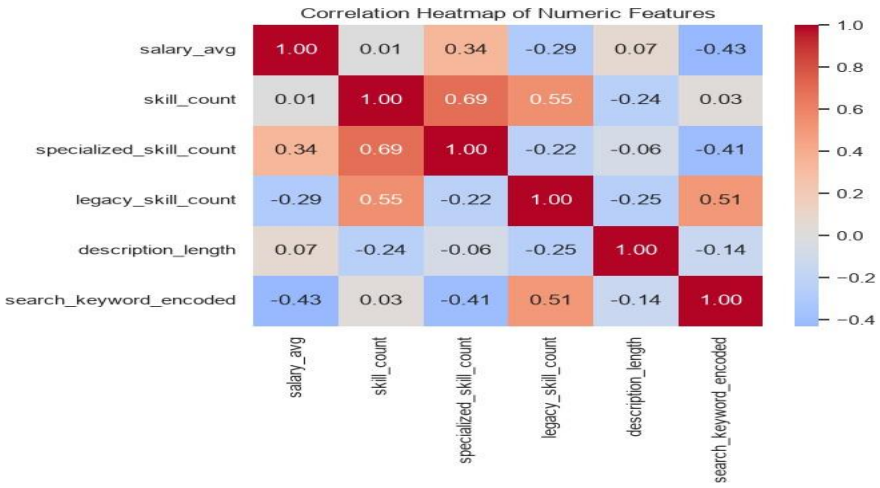


Description length vs. skill count.

Key observation: The correlation between description length and skill count is weak and slightly negative ($r = -0.24$). Nearly all postings cluster tightly between 60–90 words, regardless of tier.

Key Insight: This mostly reflects the Adzuna API's description truncation limit rather than genuine variation in posting detail. Description length is not a useful standalone feature for distinguishing tiers.

4.4: Correlation Analysis



Correlation heatmap of numeric features.

Key observation: salary_avg correlates positively with specialized_skill_count (r = 0.34) and negatively with legacy_skill_count (r = −0.29). Total skill_count shows almost no correlation with salary (r = 0.01).

Key Insight: It is which skills appear — not how many — that drives compensation. No feature pair exceeds r = 0.7, meaning the features carry largely independent information, supporting their combined use in the classification model.

4.5: Overall Tier Comparison Summary

Dimension	Specialized	Legacy
Volume	909 (53.2%)	801 (46.8%)
Avg. Disclosed Salary	₹14.79 L	₹6.56 L
Salary Disclosure Rate	15.9%	50.8%
Avg. Skill Count	2.22	2.30
Growth Over Collection Window	+82.3%	−49.7%

Overall Insight: The descriptive and relationship analysis of 1,710 postings shows a market that is volume-balanced but structurally divided: specialized roles command far higher — but less transparent — pay, use a completely distinct skill vocabulary, and are growing sharply, while legacy roles are more transparent on pay but are shrinking in relative volume.

5. Machine Learning Report

5.1 Objective

The objective of the modeling phase was to build a classification model to predict whether a job posting is “specialized” or “legacy” using only content-derived features from the posting itself.

Machine learning was used to test whether skill and description signals alone can reliably separate the two hiring tiers — directly testing the core “structural mismatch” premise of the project.

5.2 Target Variable

The target variable selected for prediction was:

tier_label — specialized or legacy

The input features used were:

- specialized_skill_count
- legacy_skill_count
- skill_count
- description_length

The search_keyword field was deliberately excluded, since tier_label was directly derived from it and including it would constitute label leakage.

5.3 Train-Test Split

The dataset was divided into:

- 80% Training Data (1,369 records)
- 20% Testing Data (341 records)

A stratified split was used to preserve the roughly even tier balance in both sets. The training data was used to build the models, while the testing data was used to evaluate performance on unseen postings.

5.4 Models Used

Because the development machine's security policy also blocked scikit-learn's compiled dependencies, both models were implemented entirely from scratch using only NumPy and pandas.

5.4.1. Logistic Regression

Logistic Regression was implemented from scratch using gradient descent, selected as the baseline model because it is simple and interpretable.

5.4.1.2 Model Evaluation

Evaluation metrics used: Accuracy, Precision, Recall, F1 Score.

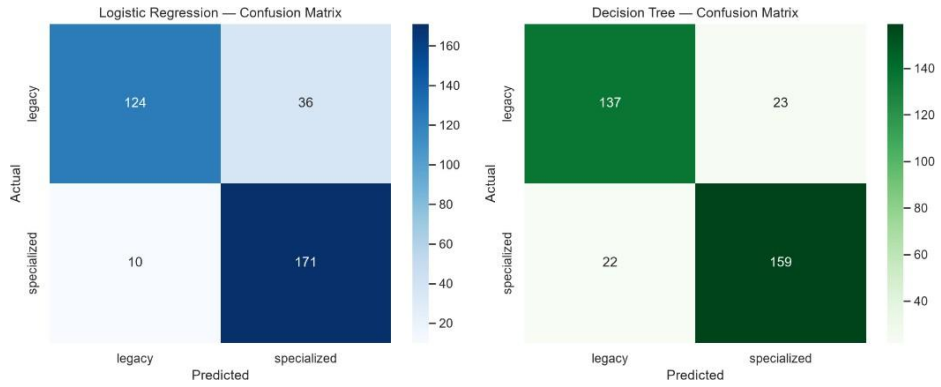
Metric	Value
Accuracy	86.5%
Precision (Specialized)	82.6%
Recall (Specialized)	94.5%

5.4.2. Decision Tree

A Decision Tree was implemented from scratch using recursive Gini-impurity splitting, selected as the second model because it can capture non-linear relationships between posting features and tier.

5.4.2.1 Model Evaluation

Metric	Value
Accuracy	86.8%
Precision (Specialized)	87.4%
Recall (Specialized)	87.8%



Confusion matrices for both models on the 341-row test set.

5.5: Model Comparison

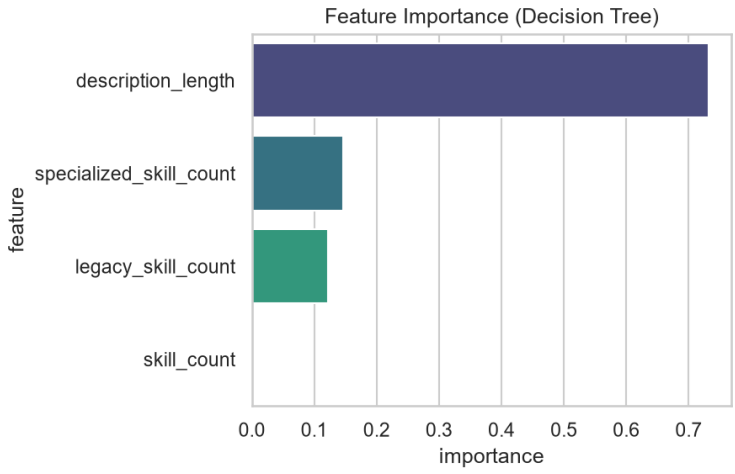
Logistic Regression

Logistic Regression achieved 86.5% accuracy, 82.6% precision and 94.5% recall for the specialized class. This indicates the model is highly sensitive — it misses very few truly specialized postings (only 10 of 181) — useful if the model were used to flag high-value roles for review.

Decision Tree

Decision Tree achieved 86.8% accuracy, with a better balance between precision (87.4%) and recall (87.8%) than Logistic Regression.

Both models substantially outperform a random baseline (~53%, given the near-even class split), confirming that specialized and legacy postings form genuinely separable segments based on content alone.



Decision Tree feature importance.

Insight: Description length emerged as the single most influential feature (the majority of the tree's splits), followed by specialized_skill_count and legacy_skill_count — indicating that beyond specific keywords, the overall structure of a posting's description carries meaningful signal about which tier it belongs to.

5.6: Best Model

Based on the evaluation results, the Decision Tree was selected as the best-performing model, achieving the highest accuracy (86.8%) with the best precision/recall balance.

Important: an accuracy of 86.8% means the model correctly classifies roughly 9 out of 10 postings — a strong result for a model built entirely from four simple, content-derived features.

5.7: Final Conclusion

The Decision Tree model performed marginally better than Logistic Regression for classifying job postings by tier. Both models' strong, well-above-baseline accuracy demonstrates that specialized and legacy postings are genuinely distinct segments of the job market, detectable from content alone — directly supporting the project's core hiring-paradox thesis.

6. Power BI Interactive Dashboard Report

6.1 Dashboard Objective

The Power BI dashboard was developed to transform the tech job market analysis into an interactive and user-friendly reporting solution. The dashboard presents job volume, salary, skill demand, hiring trends, AI-based insights, model results, and individual posting details.

The dashboard allows users to explore the specialized-vs-legacy hiring divide interactively through filters, KPIs, charts, AI visuals, and drill-through analysis.

6.2 Dashboard Pages

The dashboard contains 4 pages:

1. Executive Overview – Overall market volume, salary, and trend analysis

2. Skill Demand & AI Driver Analysis – Skill "barbell" divide and AI-driven key influencers

3. Model Audit & Evaluation – Classification model performance and feature weights

4. Role Explorer – Drill-through job-level detail and misclassification audit

6.3 Key Features

- KPI cards
- Synchronized slicers (location, salary disclosure, date)
- Trend, bar, scatter, and donut charts
- Key Influencers (AI visual)
- Row-Level Security (Specialized_Recruiter / Legacy_Recruiter roles)
- Drill-through
- Reset Filters and Model Selector bookmarks
- Interactive navigation

Insight: The dashboard provides an interactive view of the complete tech hiring paradox analysis, from executive summary through to individual job-posting detail.

6.4 Page 1 — Executive Overview

The Executive Overview provides a summary of the 1,710 job postings in the dataset.

Key Performance Indicators

- Total Postings: 1,710
- Specialized Share: 53.2% (vs. 46.8% Legacy)
- Avg. Specialized Salary: ₹14.79 L
- Avg. Legacy Salary: ₹6.56 L
- Paradox Pay Gap: +₹8.23 L (+125.6%)

The page also contains:

- Donut chart of market volume balance by tier
- Clustered bar chart: role compensation ladder (average salary by role and tier)
- Line chart: hiring trend over time by tier
- Stacked bar chart: postings by top geographic hub and tier
- Synced slicers: location, salary disclosed, month-year
- Reset Filters bookmark button

Insight: Market volume is nearly balanced between tiers, but the two segments diverge sharply on pay and growth trend — confirming the hiring paradox is about job structure, not job scarcity.

6.5: Page 2 — Skill Demand

This page dissects the skill "barbell" distribution and identifies what pushes a posting into the higher-paying tier.

Key Metric Badges

- Average Skills per Posting: 2.26
- Specialized Skills in Specialized Roles: 2.07 vs. 0.15 in Legacy
- Legacy Skills in Legacy Roles: 1.94 vs. 0.37 in Specialized
- Salary Disclosure Rate: 32.3%

The major visuals include:

- Horizontal bar chart: top 15 in-demand skills overall

- Grouped column chart: the skill “barbell” divide (specialized vs. legacy skill count by tier)
- Scatter plot: specialized skill count vs. salary, colored by tier

Insight: specialized_skill_count as the strongest single driver of a posting being classified as specialized — reinforcing the model's own feature findings.

6.6: Page 3 — Model Audit & Evaluation

This page benchmarks the from-scratch classification algorithms and analyzes accuracy and confusion areas.

Model Performance KPI Cards

- Test Set Sample Size: 341 postings
- Decision Tree Accuracy: 86.8%
- Logistic Regression Accuracy: 86.5%
- Specialized Class Recall (Logistic Regression): 94.5%

The major visuals include:

- Matrix visual: Decision Tree confusion matrix (actual_tier × predicted tier)
- Matrix visual: Logistic Regression confusion matrix
- Clustered horizontal bar chart: feature importance / coefficients
- Model Selector bookmark toggle: switch between Decision Tree and Logistic Regression views

Insight: Both models comfortably beat the ~53% random baseline, empirically confirming that specialized and legacy postings are distinguishable from content alone.

6.7 Page 4 — Role Explorer

The Role Explorer page provides a detailed, drill-through view of individual job postings, and supports auditing model errors.

Drill-through is configured on search_keyword and tier_label, so right-clicking any role/tier chart on Pages 1–3 opens this page filtered to that selection, with a Back button to return.

Dynamic Filter Badges

- Selected Role
- Filtered Postings (count)
- Avg. Role Salary
- Avg. Skill Count

The page displays:

- Main interactive table: job_title, company, location, tier_label, salary_avg, extracted_skills, predicted tier, and job_url (clickable link)
- Misclassification audit card, filtered to postings where the predicted tier did not match the actual tier — useful for reviewing hybrid postings (e.g. an IT support role that also asks for cloud tools)

Insight: Drill-through functionality allows users to move from overall dashboard analysis to detailed, individual-posting information, and to inspect exactly where the model's predictions diverge from reality.

7. Recommendations

As a Data Scientist, based on the EDA, correlation analysis, skill and salary analysis, and machine learning results, I would recommend the following actions:

1. Prioritize Specialized Upskilling for Job Seekers

The analysis shows a substantial and growing pay premium for specialized skills.

Recommendation: Entry-level and legacy-skill candidates should treat upskilling toward cloud, AI, or data-focused competencies as a near-term priority.

2. Design Reskilling as a Distinct Bridge Pathway

Specialized and legacy postings share zero overlap in top-10 required skills.

Recommendation: Reskilling programs should be built as a distinct "bridge" pathway rather than an incremental extension of legacy-role training.

3. Improve Salary Transparency for Specialized Roles

Specialized roles disclose salary far less often (15.9%) than legacy roles (50.8%).

Recommendation: Platforms and policymakers should consider transparency requirements or incentives specifically targeting high-value/specialized postings, where the pay gap is currently least visible.

4. Use Skill Identity, Not Skill Count, for Role Tagging

Skill count is nearly identical between tiers, but skill vocabulary is completely distinct.

Recommendation: Any automated job-classification or matching system should weight which specific skills appear over how many are listed.

5. Track the Specialized-vs-Legacy Hiring Trend Over Time

Specialized postings grew +82.3% while legacy postings fell −49.7% within the collection window.

Recommendation: Job platforms and workforce planners should monitor this ratio on an ongoing basis as an early indicator of market-wide skill mismatch.

Overall Business Recommendation

As a Data Scientist, I recommend using salary patterns, skill vocabulary, hiring trends, and machine learning classification together to support data-driven workforce planning, reskilling program design, and job-market transparency policy — treating tech unemployment as a structural skills-mismatch problem that available data can help address directly.

8. Limitations & Future Work

- This analysis is demand-side only (job postings). It does not include labor-supply data — unemployment rates, candidate skill inventories, or applicant volume — which would be needed to fully separate a skill-mismatch explanation from a pure jobs-shortage explanation.
- Salary comparisons are based on the 32.3% of postings that disclose pay; this subset may not be fully representative of the whole market.
- The Adzuna API truncates job descriptions to roughly 60–90 words, which may understate true skill requirements for some postings.
- The growth/decline trend reflects a short data-collection window and should be validated against a longer time series before treating it as a durable market trend.
- Future work could incorporate government labor statistics (e.g. India's Periodic Labour Force Survey) or platform-level application data to test the labor-supply side of the hypothesis directly, and extend the model with unsupervised clustering to check whether the specialized/legacy split emerges naturally from the data.

9. Appendix — Project Deliverables

- Data acquisition script (Adzuna API): webScraping.py
- Data cleaning & feature engineering script: cleaned.py
- Cleaned dataset: cleaned_jobs.csv (1,710 rows)
- EDA notebook: Eda.ipynb
- Model building notebook: Model.ipynb
- Model predictions (test set): model_predictions.csv
- Interactive Power BI dashboard: Dashboard.pbix